Time Series Analysis and Forecasting

Seminar 12: Spectral Analysis and Frequency-Domain Methods

Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Seminar Outline

- **Part I: Year 3 Level** – Fundamentals of spectral analysis
  - Exercise 1: Log-Returns and White Noise
  - Exercise 2: Raw Periodogram
  - Exercise 3: Periodogram Inconsistency
  - Exercise 4: AR(1) Spectrum
- **Part II: Master Level** – Advanced estimation methods
  - Exercise 5: Welch vs Multitaper
  - Exercise 6: AR Spectral Estimation
  - Exercise 7: HP Filter Transfer Function
- **Part III: PhD Extension** – Long memory and coherence
  - Exercise 8: GPH Estimator
  - Exercise 9: Local Whittle
  - Exercise 10: Cross-Spectral Analysis

Exercise 1: Log-Returns

Question

Given a price series P_t, define log-returns $r_t = \log(P_t/P_{t-1})$.

Exercise

(a) Plot the price series P_t and the return series r_t. What visual features do you notice?

(b) Compute the sample ACF of r_t and of r_t^2 for lags $h = 1, \ldots, 20$.

(c) What does a near-zero ACF of r_t suggest about predictability? What do the significant ACF values of r_t^2 indicate?

Answer on next slide...

Exercise 1: Solution

Solution

(a) P_t typically shows a non-stationary trending pattern; r_t fluctuates around zero with periods of high and low volatility (volatility clustering).

(b) Sample ACF of r_t: most values fall within $\pm 1.96/\sqrt{n}$ confidence bands $\Rightarrow$ returns are approximately uncorrelated. Sample ACF of r_t^2: significant positive values at many lags $\Rightarrow$ squared returns are correlated.

(c)
- Near-zero ACF(r_t) $\Rightarrow$ returns are close to **white noise**: past returns carry little information to forecast future returns (consistent with weak-form market efficiency).
- Significant ACF(r_t^2) $\Rightarrow$ **ARCH/GARCH effects**: volatility is predictable even when returns are not.
- **Key insight**: a flat spectral density of r_t implies equal power at all frequencies — the hallmark of white noise.

TSA_ch12_periodogram

Exercise 2: Raw Periodogram

Question

Given $n = 256$ observations, the periodogram at Fourier frequency $\omega_j = 2\pi j/n$ is:

$$I(\omega_j) = \frac{1}{2\pi n} |d(\omega_j)|^2, \qquad d(\omega_j) = \sum_{t=1}^{n} Y_t \, e^{-i\omega_j t}$$

Exercise

(a) Plot $I(\omega_j)$ against $\omega_j \in [0, \pi]$. How do you identify the dominant frequency?

(b) If the series has a *quarterly* pattern (period $= 4$), at which angular frequency ω^* do you expect the largest periodogram peak?

(c) What does a completely flat periodogram indicate?

Answer on next slide...

Exercise 2: Solution

Solution

(a) The dominant frequency is the ω_j at which $I(\omega_j)$ reaches its global maximum. The corresponding period is $T = 2\pi/\omega_j$ time units.

(b) A quarterly pattern has period $T = 4$, so:

$$\omega^* = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2} \approx 1.571 \text{ rad}$$

This is the 64th Fourier frequency for $n = 256$ (since $j^* = n/T = 64$).

(c) A flat periodogram $I(\omega_j) \approx c$ for all j indicates **white noise**: all frequencies contribute equally to the series variance ($\int_0^\pi f(\omega)\, d\omega = \gamma(0)$, where $f(\omega) = \sigma^2/(2\pi) = \text{const}$).

TSA_ch12_periodogram

Exercise 3: Periodogram Inconsistency

Question

Show that the periodogram is an *inconsistent* estimator of the spectral density.

Exercise

(a) State the asymptotic distribution of $I(\omega_j)$ as $n \to \infty$.

(b) Derive $\mathrm{Var}[I(\omega_j)]$ as a function of n. Does it go to zero?

(c) Propose a solution that produces a consistent estimator, and explain the bias-variance tradeoff involved.

Answer on next slide...

Exercise 3: Solution

Solution

(a) For $\omega_j \neq 0, \pi$: $\dfrac{2\,I(\omega_j)}{f(\omega_j)} \xrightarrow{d} \chi^2_2 \;\Rightarrow\; I(\omega_j) \overset{d}{\approx} f(\omega_j) \cdot \chi^2_2/2$

(b) Since χ^2_2 has variance 4: $\mathrm{Var}[I(\omega_j)] \approx f(\omega_j)^2 \cdot 4/4 = f(\omega_j)^2$. This does **not** decrease with $n \Rightarrow$ periodogram is **inconsistent**.

(c) **Smoothing** over m adjacent ordinates:

$$\hat{f}(\omega_j) = \frac{1}{2m+1} \sum_{k=-m}^{m} I(\omega_{j+k})$$

Variance $\propto 1/m$, bandwidth $\propto m/n$. More smoothing $\Rightarrow$ lower variance but higher bias.

TSA_ch12_periodogram

Exercise 4: AR(1) Spectrum

Question

Consider the AR(1) process $Y_t = 0.8\, Y_{t-1} + \varepsilon_t$ with $\sigma_\varepsilon^2 = 1$.

Exercise

(a) Derive the spectral density $f(\omega)$ for $\omega \in [0, \pi]$.

(b) Compute the ratio $f(0)/f(\pi)$. Interpret the result.

(c) What is a *red spectrum*? Give an economic example.

Answer on next slide...

Exercise 4: Solution

Solution

(a) For AR(1) with coefficient $\phi = 0.8$:

$$f(\omega) = \frac{\sigma^2}{2\pi} \cdot \frac{1}{1 - 2\phi\cos\omega + \phi^2} = \frac{1}{2\pi} \cdot \frac{1}{1.64 - 1.6\cos\omega}$$

(b) At $\omega = 0$: $\cos 0 = 1 \Rightarrow f(0) = \frac{1}{2\pi} \cdot \frac{1}{(1-\phi)^2} = \frac{1}{2\pi} \cdot \frac{1}{0.04} = \frac{25}{2\pi}$.

At $\omega = \pi$: $\cos\pi = -1 \Rightarrow f(\pi) = \frac{1}{2\pi} \cdot \frac{1}{(1+\phi)^2} = \frac{1}{2\pi} \cdot \frac{1}{3.24}$.

$$\frac{f(0)}{f(\pi)} = \left(\frac{1+\phi}{1-\phi}\right)^2 = \left(\frac{1.8}{0.2}\right)^2 = 81$$

(c) A **red spectrum** is dominated by low frequencies: most variance is at slow oscillations (long cycles). GDP, consumption, and stock prices all exhibit red spectra — they move slowly and trend persistently, with little energy at high (business-cycle) frequencies.

TSA_ch12_spectral_density

Exercise 5: Welch vs. Multitaper Estimation

Question

Compare two spectral estimators on the same series of $n = 512$ observations:

- **Welch**: $L = 4$ overlapping segments
- **Multitaper**: $NW = 4$, $K = 7$ DPSS tapers

Exercise

(a) Compute the frequency resolution (bandwidth) for each estimator.

(b) Which estimator has lower variance? Why?

(c) Describe the bias-variance tradeoff for each method.

Answer on next slide...

Exercise 5: Solution

Solution

(a) Welch ($L = 4$, $n_{seg} = 128$): $\Delta\omega_{\text{Welch}} \approx 2/128 \approx 0.0156$ cyc/obs.
Multitaper ($NW = 4$, $n = 512$): $\Delta\omega_{\text{MT}} \approx 2NW/n = 8/512 = 0.0156$ cyc/obs.
Both have the same frequency resolution for these parameters.

(b) Multitaper averages $K = 7$ independent eigen-spectra $\Rightarrow$ variance $\propto 1/K$. For equal bandwidth, multitaper has lower variance: equivalent degrees of freedom $\approx 2K = 14$ vs $\approx 2L = 8$.

(c) Both methods trade frequency resolution for variance reduction. Multitaper achieves near-optimal sidelobe suppression via DPSS tapers, minimising spectral leakage — preferred for most applications.

TSA_ch12_estimation

Exercise 6: AR Spectral Estimation

Question

Consider the AR(2) process:

$$Y_t = 1.5\, Y_{t-1} - 0.7\, Y_{t-2} + \varepsilon_t, \qquad \sigma_\varepsilon^2 = 1$$

Exercise

(a) Derive the parametric spectral density $\hat{f}_{AR}(\omega)$.

(b) Find the peak frequency ω^* of the spectrum analytically.

(c) Convert ω^* to a cycle period and interpret economically.

Answer on next slide...

Exercise 6: Solution

Solution

(a) With $\phi_1 = 1.5$, $\phi_2 = -0.7$ and transfer function $A(e^{-i\omega}) = 1 - 1.5e^{-i\omega} + 0.7e^{-2i\omega}$:

$$f(\omega) = \frac{\sigma^2}{2\pi\,|A(e^{-i\omega})|^2} = \frac{1}{2\pi\,|1 - 1.5e^{-i\omega} + 0.7e^{-2i\omega}|^2}$$

(b) The spectral peak for an AR(2) with complex roots occurs at:

$$\omega^* = \arccos\left(\frac{\phi_1}{2|\phi_2|} \cdot \frac{|\phi_2| + 1}{1}\right) = \arccos\left(\frac{1.5}{2 \times 0.7}\right) = \arccos(1.071)$$

Since $\arccos(1.071)$ has no real solution, we differentiate $|A|^2$ numerically: $\omega^* \approx 0.47$ rad.

(c) Period $= 2\pi/\omega^* \approx 2\pi/0.47 \approx 13.4$ observations. For quarterly data, this corresponds to a **3.3-year business cycle** — consistent with typical inventory and investment cycles.

TSA_ch12_spectral_density

Exercise 7: HP Filter Transfer Function

Question

The HP filter extracts trend τ_t by minimising $\sum(Y_t - \tau_t)^2 + \lambda \sum(\Delta^2 \tau_t)^2$. The squared gain of the *cycle* component is:

$$|A_{HP}(\omega)|^2 = \frac{\lambda\,[2(1-\cos\omega)]^2}{1+\lambda\,[2(1-\cos\omega)]^2}$$

Exercise

(a) Evaluate $|A_{HP}(\omega)|^2$ at $\omega = 0$ and $\omega = \pi$ for any λ.

(b) Find the half-power cutoff frequency for $\lambda = 1600$.

(c) Summarise Hamilton (2018)'s critique of the HP filter.

Answer on next slide...

Exercise 7: Solution

Solution

(a) At $\omega = 0$: $2(1 - \cos 0) = 0 \Rightarrow |A_{HP}(0)|^2 = 0$ (trend removed).
At $\omega = \pi$: $2(1 + 1) = 4 \Rightarrow |A_{HP}(\pi)|^2 = 16\lambda/(1 + 16\lambda) \approx 1$ (high freq. passed).

(b) Setting $|A_{HP}(\omega)|^2 = 0.5$: $\lambda[2(1 - \cos\omega)]^2 = 1$, so $2(1 - \cos\omega) = 1600^{-1/2} = 0.025$, giving $\cos\omega \approx 0.9875$, $\omega^* \approx 0.158$ rad $\approx 2\pi/40$ (period ≈ 40 quarters $= 10$ years).

(c) **Hamilton (2018)**: HP filtering is equivalent to a regression on lagged values. The resulting cycle is spuriously correlated at all lags and imposes ARMA(4,4) structure on white noise — generating *artificial* business cycles. Hamilton recommends a simple 2-step regression alternative.

TSA_ch12_filters

Exercise 8: GPH Estimator

Question

For ARFIMA(0,d,0), the GPH estimator fits OLS to:

$$\log I(\omega_j) = c - d \log[4 \sin^2(\omega_j/2)] + u_j, \quad j = 1, \ldots, m$$

exploiting the approximation $\log f(\omega) \approx c - d \log[4 \sin^2(\omega/2)]$ near $\omega = 0$.

Exercise

(a) What is the OLS estimator $\hat{d}$? How is it computed?

(b) What bandwidth m should be used? Theoretical recommendations?

(c) What is the asymptotic standard error of $\hat{d}$?

Answer on next slide...

Exercise 8: Solution

Solution

(a) Let $x_j = -\log[4\sin^2(\omega_j/2)]$ and $y_j = \log I(\omega_j)$:

$$\hat{d} = \frac{\sum_{j=1}^{m}(x_j - \bar{x})(y_j - \bar{y})}{\sum_{j=1}^{m}(x_j - \bar{x})^2} \quad \text{(OLS slope)}$$

(b) Theory: $m = O(n^\alpha)$, $\alpha \in (0, 1)$. Common practice: $m \approx n^{0.5}$ to $n^{0.6}$. Too small $m \Rightarrow$ high variance; too large $m \Rightarrow$ bias.

(c) $\mathrm{SE}(\hat{d}) = \pi/\sqrt{6m}$. For $n = 1000$, $m = 32$: SE ≈ 0.226. Hurst: $H = d + 0.5$; persistent series have $d > 0$ ($H > 0.5$).

TSA_ch12_long_memory

Exercise 9: Local Whittle Estimator

Question

Compare the GPH and Local Whittle estimators for a series of $n = 1000$ observations with true $d = 0.3$.

Exercise

(a) Write the Local Whittle objective function and state what it minimises.

(b) What bandwidth m is used, and what is the asymptotic variance of $\hat{d}_{LW}$?

(c) Why is Local Whittle more efficient than GPH?

Answer on next slide...

Exercise 9: Solution

Solution

(a) Local Whittle minimises the restricted Whittle log-likelihood:

$$Q(d) = \log G(d) - 2d\,\bar{\lambda}_m, \quad G(d) = \frac{1}{m}\sum_{j=1}^{m} \frac{I(\omega_j)}{\omega_j^{-2d}}, \quad \hat{d}_{LW} = \arg\min_d Q(d)$$

(b) Bandwidth $m = O(n^{0.6\text{–}0.8})$; for $n = 1000$ typically $m \approx 100$–250. Asymptotic variance: $\mathrm{Var}(\hat{d}_{LW}) = 1/(4m)$. For $m = 100$: SD $= 0.05$.

(c) GPH uses OLS on log-periodogram (heavy-tailed χ^2_2 errors); LW uses a proper likelihood approximation. LW is asymptotically efficient: $\mathrm{Var} = 1/(4m)$ vs GPH's $\approx 1.645/m$ for the same m.

TSA_ch12_long_memory

Exercise 10: Cross-Spectral Analysis

Question

Suppose $Y_t = X_{t-4} + \varepsilon_t$ where $\varepsilon_t \perp X_s$ for all s, t.

Exercise

(a) Derive the cross-spectrum $S_{XY}(\omega) = e^{-4i\omega} S_{XX}(\omega)$. What is the phase function $\phi(\omega)$?

(b) Compute the squared coherence $C^2_{XY}(\omega)$ (i) without noise, (ii) with noise $\varepsilon_t \sim (0, \sigma^2_\varepsilon)$.

(c) What does the phase slope $-d\phi/d\omega$ tell you, and what is its value here?

Answer on next slide...

Exercise 10: Solution

Solution

(a) Taking the Fourier transform of $E[Y_t X^*_{t-h}]$: $Y_t = X_{t-4} + \varepsilon_t$ gives the transfer function $H(\omega) = e^{-4i\omega}$, so:

$$S_{XY}(\omega) = H(\omega)\, S_{XX}(\omega) = e^{-4i\omega}\, S_{XX}(\omega)$$

The phase is $\phi(\omega) = -4\omega$ (linear in ω).

(b) Squared coherence $C^2_{XY}(\omega) = |S_{XY}|^2/(S_{XX} \cdot S_{YY})$:

(i) No noise: $S_{YY} = S_{XX} \Rightarrow C^2_{XY}(\omega) = 1$ at all frequencies (perfect linear relationship).
(ii) With noise: $S_{YY} = S_{XX} + \sigma^2_\varepsilon/(2\pi) \Rightarrow C^2_{XY}(\omega) = S_{XX}(\omega)/[S_{XX}(\omega) + \sigma^2_\varepsilon/(2\pi)] < 1$.

(c) The **group delay** $= -d\phi/d\omega$ measures the frequency-specific time lag:

$$-\frac{d\phi}{d\omega} = -\frac{d(-4\omega)}{d\omega} = 4 \text{ periods at ALL frequencies}$$

A constant group delay confirms the system introduces a **pure shift of 4 periods** with no phase distortion.

TSA_ch12_coherence

Summary: Chapter 12

Key Concepts by Level

Part	Exercises	Key Concepts
Year 3	Ex. 1–4	Log-returns; flat ACF $\Rightarrow$ white noise; periodogram definition; inconsistency; AR(1) spectrum; red vs blue spectrum
Master	Ex. 5–7	Welch vs multitaper bias-variance tradeoff; AR parametric spectrum; HP filter transfer function; Hamilton critique
PhD	Ex. 8–10	GPH log-periodogram regression; Local Whittle efficiency; cross-spectrum; squared coherence; group delay

The Fundamental Tradeoff

Frequency resolution $\longleftrightarrow$ **Statistical variance**

More smoothing $\Rightarrow$ lower variance but cannot resolve nearby spectral peaks

Questions?

Bibliography

Core References

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Long Memory & Filters

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Code: `https://github.com/danpele/TSA/tree/main/TSA_ch12` **Quantlet**: `https://quantlet.com` **Quantinar**: `https://quantinar.com`

Thank You!

Questions?

Seminar materials are available at: https://danpele.github.io/Time-Series-Analysis/

Quantlet Quantinar